

From: quality.assurance@westpharma.com
To: pharma.procurement@pfizer.com
Date: 12 February 2025
Subject: SDF Package - West Item 4432/50 Grey FC (FluroTec + B2-Coating)

----- Forwarded message -----

Dear Pfizer Procurement Team,

Please find attached the complete Supplier Documentation File (SDF) for West Pharmaceutical Services Item 4432/50 Grey FC, a 20 mm chlorobutyl rubber stopper with FluroTec barrier film and B2-coating, supplied for use in injectable drug product primary packaging.

This package covers the following batches released in Q4 2024:

D000291847 | D000291852 | D000291863 | D000291871 | D000291885

Documents included in this SDF package:

1. Quality Certificate No. 80372784-000010 (3 pages)
2. Packaging Component Specification - WPS-SPEC-4432-REV-D
3. Animal Origin Statement - WPS-AOS-2024-001
4. Compliance Bulletin CB-4432-050-FCB2 Rev 5
5. Supplier Qualification Record SUP-WPS-4432-2024
6. STERIS Certificate of Processing GTN-2025-0183A

All documents are current and comply with ICH Q7, EU GMP Annex 15, and 21 CFR Part 211 requirements for primary packaging components.

Please do not hesitate to contact our Quality team if you require any additional information or clarification.

Regards,
Melissa Hartmann
Senior Quality Assurance Manager
West Pharmaceutical Services, Inc.

West Pharmaceutical Services, Inc.

347 Oliver Street, Jersey Shore, PA 17740
Tel: +1 (570) 398-5200 | www.westpharma.com

QUALITY CERTIFICATE

No. 80372784-000010

Certificate Date:	18 December 2024
Customer:	Pfizer Manufacturing Belgium NV
Customer PO:	4500917432
West Item No.:	4432/50 Grey FC
Description:	20mm Chlorobutyl Rubber Stopper, FluroTec + B2-Coating
Drawing No.:	DWG-4432-050-FCB2-R4

Section 1 - Compliance Statements

West Pharmaceutical Services, Inc. hereby certifies that the above-referenced product has been manufactured, tested, and released in accordance with the applicable West Pharmaceutical Services quality specifications and the purchase order requirements specified by the customer.

This product complies with the following regulatory and pharmacopoeial standards:

- * USP <381> Elastomeric Closures for Injections
- * EP 3.2.9 Rubber Closures for Containers for Aqueous Parenteral Preparations
- * JP 7.03 Rubber Closure Test for Aqueous Infusions
- * ISO 8362-5:2016 Rubber Closures for Injection Vials
- * 21 CFR 177.2600 Rubber Articles Intended for Repeated Use

FluroTec® barrier film applied to stopper contact surface is manufactured from modified polytetrafluoroethylene (PTFE) resin per West proprietary specification WPS-FT-RESIN-001 Rev C.

QUALITY CERTIFICATE No. 80372784-000010 (continued)

West Item: 4432/50 Grey FC | Page 2 of 3

Section 2 - Batch / Lot Identification

Batch No.	Qty (k pcs)	Mfg Date	Exp Date	Release Date	Site
D000291847	500	2024-10-02	2029-10-01	2024-10-21	Lionville, PA
D000291852	500	2024-10-08	2029-10-07	2024-10-28	Lionville, PA
D000291863	500	2024-10-15	2029-10-14	2024-11-04	Lionville, PA
D000291871	500	2024-10-22	2029-10-21	2024-11-11	Lionville, PA
D000291885	500	2024-10-30	2029-10-29	2024-11-19	Lionville, PA

Section 2a - Traceability

Each batch is manufactured from a single compound lot. Compound lot numbers are traceable to raw material certificates of analysis retained on file at the manufacturing site (Lionville, PA) for a minimum of 15 years per 21 CFR 211.180.

Batch No.	Compound Lot	Elastomer Raw Material	COA Reference
D000291847	CMP-24-C-00412	Chlorobutyl Rubber, Grade 1066	COA-4412-2024-09
D000291852	CMP-24-C-00412	Chlorobutyl Rubber, Grade 1066	COA-4412-2024-09
D000291863	CMP-24-C-00419	Chlorobutyl Rubber, Grade 1066	COA-4419-2024-10
D000291871	CMP-24-C-00419	Chlorobutyl Rubber, Grade 1066	COA-4419-2024-10
D000291885	CMP-24-C-00427	Chlorobutyl Rubber, Grade 1066	COA-4427-2024-10

QUALITY CERTIFICATE No. 80372784-000010 (continued)

West Item: 4432/50 Grey FC | Page 3 of 3

Section 3 - Analysis / Test Results

Test Parameter	Specification	D847	D852	D863	D871	D885
Appearance	No visible defects	Pass	Pass	Pass	Pass	Pass
Dimensions (height, mm)	6.80 ± 0.30	6.82	6.81	6.79	6.83	6.80
Shore A Hardness	50 ± 5	52	51	50	53	52
Residue on Ignition (%)	<= 1.0	0.42	0.44	0.41	0.43	0.42
Fragmentation (pcs)	<= 10	2	1	3	2	2
Self-Sealing (passes)	>= 10	10	10	10	10	10
Penetration Force (N)	10 - 50	22	24	21	23	22
Extractables pH	5.0 - 7.5	6.8	6.9	6.7	6.8	6.8
FluroTec Adhesion	Conforms	Conf	Conf	Conf	Conf	Conf
Particulate Matter	Per USP <381>	Pass	Pass	Pass	Pass	Pass
Sterility (final pkg)	Sterile	Pass	Pass	Pass	Pass	Pass

Authorised Release Statement

All batches listed above have been reviewed against applicable specifications and are hereby released for use in parenteral drug product manufacturing.

Authorised by: Director, Analytical Sciences & Quality Release
West Pharmaceutical Services, Inc., Lionville, PA 19341
Date: 18 December 2024

This document has been electronically signed and is valid without a wet signature.

West Pharmaceutical Services, Inc.

347 Oliver Street, Jersey Shore, PA 17740
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Packaging Component Specification

Document No.: WPS-SPEC-4432-REV-D
Revision: D
Effective Date: 2024-01-15
Product: 4432/50 Grey FC - 20mm Chlorobutyl Stopper
Drawing Ref.: DWG-4432-PKG-R6

1. Scope

This specification defines the primary and secondary packaging configuration for Item 4432/50 Grey FC rubber closures as supplied to Pfizer Manufacturing Belgium NV for use in parenteral vial filling operations.

2. Primary Packaging

Component	Material / Grade	Specification / Standard
Stopper Bag	Polyethylene (LDPE)	WPS-MAT-PE-001, 0.08 mm min wall
Bag Closure	Cable tie + heat seal	Hermetic seal, 5 kPa burst
Desiccant Sachet	Silica gel, 10 g	Per MIL-D-3464E

3. Secondary Packaging

Component	Material / Grade	Specification
Outer Carton	Corrugated cardboard	B-flute, 32 ECT min
Packing Density	5,000 pcs per bag	Per DWG-4432-PKG-R6
Bags per Carton	10 bags (50,000 pcs)	Max gross wt: 18 kg

4. Revision History

Rev	Date	ECN No.	Description of Change
A	2017-03-10	ECN-2017-0041	Initial release
B	2019-08-22	ECN-2019-0183	Added desiccant sachet (humidity control)
C	2022-05-04	ECN-2022-0309	Increased packing density 4k to 5k per bag
D	2024-01-15	ECN-2024-0027	Updated carton ECT rating (28 to 32 min)

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Animal Origin Statement

Document No.: WPS-AOS-2024-001 | Rev: B | Date: 05 March 2024

Manufacturer: West Pharmaceutical Services, Inc.
Address: 347 Oliver Street, Jersey Shore, PA 17740, USA
Product Family: FluroTec-Coated Chlorobutyl Rubber Closures
Applicable Items: 4432/50 Grey FC, 4432/50 Grey FC S, 4432/20 Grey FC

Statement

West Pharmaceutical Services, Inc. hereby confirms that the above-referenced rubber closure products do not contain any materials of animal or human origin, including but not limited to:

- * Bovine-, ovine-, caprine-, or porcine-derived substances
- * Materials of ruminant origin as defined by the EU Scientific Committee
- * Human serum albumin or other human-derived excipients
- * Animal-derived processing aids retained in the final product

FluroTec® barrier film resin (modified PTFE) is of fully synthetic origin. B2-coating is a proprietary silicon-based lubricant coating; no animal-derived lubricants are used at any stage of manufacture. All raw material suppliers have confirmed in writing that their materials are free of animal-origin components.

Regulatory Basis

This statement is consistent with and addresses the requirements of:

- * European Pharmacopoeia 5.2.8 (TSE Risk Assessment)
- * EMEA/410/01 Rev 3 (Note for Guidance on Minimising TSE Risk)
- * ICH Q5A - Viral Safety Evaluation
- * FDA Guidance for Industry: Use of Materials of Animal Origin

Assessment Reference: WPS-TSE-RA-4432-FC-2024
Valid Until: 05 March 2029
Signed: Vice President, Quality & Regulatory Affairs
Date: 05 March 2024

West Pharmaceutical Services, Inc.

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Tel: +1 (570) 398-5200 | www.westpharma.com

Compliance Bulletin

CB-4432-050-FCB2 Rev 5

West Item 4432/50 Grey FC (FluroTec® + B2-Coating) | Issued: 10 January 2025

Table of Contents

1. Product Identification and Description
2. Chemical Composition
3. Physical Properties
4. Regulatory Compliance
5. Sterilization Information
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1. Product Identification and Description

West Item 4432/50 Grey FC is a 20 mm diameter chlorobutyl rubber stopper for injectable drug product vials. The stopper features a FluroTec® PTFE barrier film on the drug-contact surface and a B2 silicon-based coating on the machine-contact surface to improve handling and filling line performance.

2. Chemical Composition

Base Elastomer: Chlorobutyl rubber (CIIR), synthetic origin
Curing System: Zinc oxide, stearic acid (no elemental sulfur)
Filler: Carbon black (furnace grade, ASTM N990)
FluroTec Film: Modified PTFE (polytetrafluoroethylene), 100% synthetic
B2-Coating: Dimethicone (polydimethylsiloxane), food/pharma grade
No heavy metals, phthalates, latex, or animal-derived components.

3. Physical Properties

Shore A Hardness: 50 ± 5
Specific Gravity: $1.38 \pm 0.05 \text{ g/cm}^3$
Operating Temperature: -40°C to $+121^\circ\text{C}$
Autoclave Compatibility: 121°C , 30 min (validated)
Gamma Irradiation: Compatible up to 50 kGy

4. Regulatory Compliance

Complies with USP <381>, EP 3.2.9, JP 7.03, ISO 8362-5, and 21 CFR 177.2600.

Vendor Qualification Record

Vendor Name: West Pharmaceutical Services, Inc.
Vendor Address: 347 Oliver Street, Jersey Shore, PA 17740, USA
Vendor Code: VND-WPS-4432-2019
Qualification Status: Approved - Active Supplier
Approved For: Primary Packaging Components (Rubber Closures)

1. Qualification History

Initial Qualification Date: 22 April 2019
Re-qualification Date: 15 September 2022
Last On-Site Audit: 03 October 2024
Audit Result: Satisfactory (1 minor observation, CAPA closed)
Next Scheduled Audit: October 2026

2. Regulatory & Quality Certifications

Certification	Certificate Number	Issue Date	Expiry
ISO 9001:2015	QMS-US-2022-WPS-001	2022-09-15	2025-09-14
ISO 15378:2017	PPC-US-2022-WPS-004	2022-09-15	2025-09-14
FDA Drug Master	DMF #14738	2004-01-10	Active
EU CEP	P-2003-0149-2-0	2020-04-01	Active
ISO 14001:2015	ENV-US-2022-WPS-007	2022-09-15	2025-09-14

3. Approved Product List

West Item No.	Description	Category
4432/50 Grey FC	20mm Chlorobutyl Stopper, FluroTec + B2	Vial Closure
4432/50 Grey FC S	20mm Chlorobutyl Stopper, FluroTec + B2, Std	Vial Closure
1888/50 Grey B2	20mm Bromobutyl Stopper, B2-Coating	Vial Closure
6720/50 Grey FC	13mm Chlorobutyl Stopper, FluroTec	Vial Closure

Vendor Qualification Record (continued)

Vendor: West Pharmaceutical Services (VND-WPS-4432-2019) | Page 2 of 2

4. Manufacturing & Supply Chain

Manufacturing Site: West Pharma Lionville Facility, Lionville, PA 19341
Secondary Site: West Pharma Eschweiler Facility, Eschweiler, Germany
Distribution Center: West Pharma Distribution, Swedesboro, NJ 08085

Key Sub-Suppliers:

Sub-Supplier	Material / Service	Qualification Status
ExxonMobil Chemical	Chlorobutyl rubber (CIIR)	Approved (2019)
Daikin Industries	PTFE resin for FluroTec film	Approved (2021)
Dow Corning	Dimethicone (B2-coating base)	Approved (2019)
Cabot Corporation	Carbon black (N990 filler)	Approved (2020)
STERIS AST	Gamma irradiation processing	Approved (2019)

5. Quality Agreement

Quality Agreement Ref.: QA-WPS-PFIZER-2019-0312
Agreement Effective: 22 April 2019
Last Reviewed: 03 October 2024
Next Review Due: October 2026

Key provisions: Annual audit rights retained by Pfizer; change notification required 90 days prior to any manufacturing process or material change; CAPA responses within 30 business days for critical findings.

6. Performance Metrics (12-Month Rolling)

On-Time Delivery: 98.7%
Incoming Quality Rate: 99.9% (0 critical NCRs; 1 minor NCR, CAPA closed)
CAPA Mean Close Time: 18 business days
Customer Complaints: 0 (last 12 months)

STERIS Applied Sterilization Technologies

5960 Heisley Road, Mentor, OH 44060
Tel: +1 (440) 354-2600 | www.steris-ast.com

CERTIFICATE OF PROCESSING
Gamma Radiation Sterilization / Dose Audit

Process Run ID: GTN-2025-0183A
Process Date: 08 January 2025
Facility: STERIS AST, Mentor Gamma Facility, Mentor, OH 44060
Prepared For: West Pharmaceutical Services, Inc., Lionville, PA
Customer Ref.: WPS-PO-2024-GA-00183
Product: 4432/50 Grey FC Rubber Stoppers (20mm)
Sterilization Mode: Gamma Radiation (Co-60 Source)
Validated Dose Range: 25.0 kGy (min) - 50.0 kGy (max)

Product Lot Table

Batch No.	Qty (k pcs)	Pallet ID	Min Dose (kGy)	Max Dose (kGy)	Result
D000291847	500	PLT-0183-01	27.3	38.1	PASS
D000291852	500	PLT-0183-02	26.9	37.8	PASS
D000291863	500	PLT-0183-03	27.5	38.4	PASS
D000291871	500	PLT-0183-04	27.1	38.0	PASS
D000291885	500	PLT-0183-05	26.8	37.6	PASS

Dosimetry Summary

Dosimeters Used: Harwell Amber (HW-Amber-1010-25)
Dosimeter Traceability: NIST-traceable calibration, Cal. Date: 2024-11-01
Number of Dosimeters: 30 (6 per pallet - corners + centre)
Overall Min Dose: 26.8 kGy (Pallet PLT-0183-05, Position B-1)
Overall Max Dose: 38.4 kGy (Pallet PLT-0183-03, Position D-6)
Dose Uniformity Ratio: 1.43 (max/min; specification <= 2.50)

Signature Manifest

Role	Name / Title	Date
Processed By:	J. Kowalski, Radiation Technician	08 Jan 2025
Reviewed By:	S. Patel, Process Physicist	09 Jan 2025
Authorised By:	T. Nguyen, Quality Manager (STERIS Mentor)	09 Jan 2025